4 **(a)** By reference to energy, explain what is meant by *electromotive force* (e.m.f.).

.....

.....

..... [1]

**(b)** A 'potato cell' is formed by inserting a copper plate and a zinc plate into a potato. The circuit shown in Figure 4.1 is used in an investigation to determine the e.m.f. and internal resistance of the potato cell.

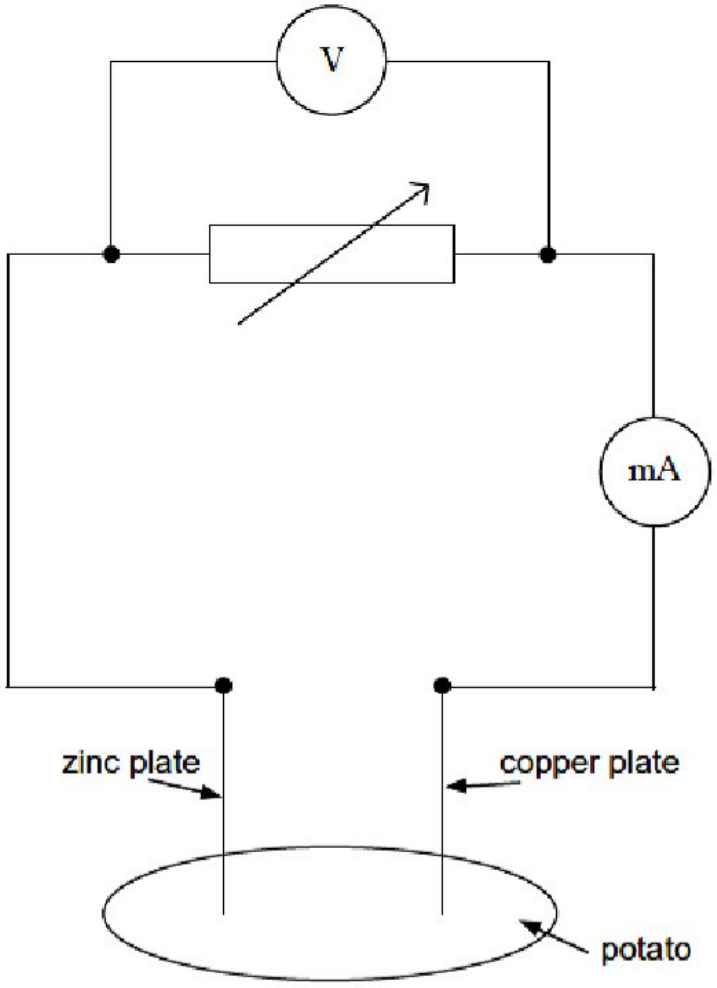



**Figure 4.1**

The plotted points × on Figure 4.2 show the data for current and voltage that were obtained by varying the resistance of the variable resistor. The line of best fit was then drawn through the plotted points.

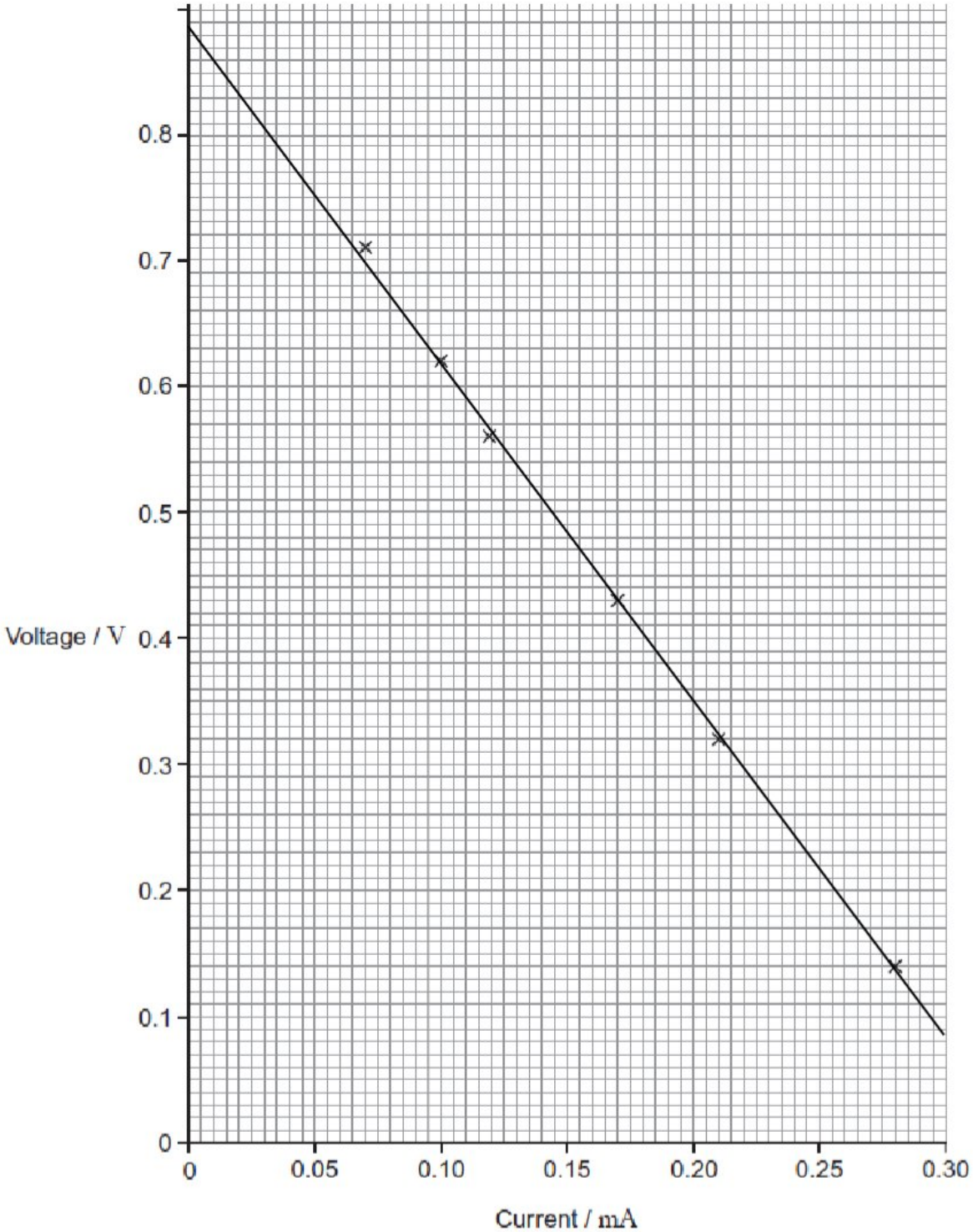



**Figure 4.2**

**(i)** State the e.m.f. of the potato cell.

e.m.f. = .....V [1]

**(ii)** Determine the internal resistance of the potato cell.

internal resistance = ……Ω [3]

**(iii)** Determine the voltage across the variable resistor when the power delivered to it is maximum. Explain your working.

voltage = ……V [3]

**(iv)** Explain why the voltages of the plotted points × on Figure 4.2 are always less than the e.m.f. of the potato cell.

……

……

……

……

……

……

……

…… [3]

**(v)** In the space below, draw a circuit diagram to suggest a modification to the circuit in Figure 4.1 which allows the e.m.f. value of the potato cell to be read directly from the voltmeter.

[1]

[Total: 12]